Victoria Knapp Perez

(949)-562-3106 | Irvine, California, USA | 1victoriakp@gmail.com | LinkedIn | GitHub | Personal Website

Summary

Physics Ph.D. candidate with a strong background in machine learning, reinforcement learning, and data-driven modeling. Experienced in AI methods such as reinforcement learning (OpenAI Gym), deep learning (PyTorch), and high-performance computing, with an emphasis on automated scientific model discovery and symbolic reasoning. Proven record of translating complex theoretical problems into computationally efficient solutions.

Education

- **University of California, Irvine**, **USA,** Ph.D. Physics (GPA=4.00) **(Expected) June 2026**
- **University of California, Irvine**, **USA,** M.S. Physics (GPA=4.00) **2023**
- **National Autonomous University of Mexico, Mexico,** B.Sc. Physics (GPA=9.88/10), Cum Laude **2020**

Skills

- Programming & Platforms: Python (7+ years), Java (1+ year), Bash scripting (5+ years), LaTeX (9+ years), Slurm (2+ years), CUDA, and High-Performance Computing (HPC) environments.
- Frameworks and Libraries: PyTorch, FastAI, Hugging Face Transformers, OpenAI Gym, scikit-learn, Pandas, NumPy, SymPy, SageMath, spaCy, TorchText, TorchVision.
- Machine Learning & Data Science: Reinforcement learning (PPO), deep learning (CNNs, RNNs, transformers, GNNs), attention and sequence-to-sequence architectures, and fine-tuning of pretrained models such as BERT, GPT, and ResNet.
- Development Tools & Version Control: Git, Jupyter Notebook, Linux shell scripting.
- Soft skills: Strong communication, collaboration, and ability to perform under pressure.
- Certifications: Coursera Java (2020).

Work Experience

University of California: Irvine, USA **2021-Present**

Research Assistant & Teaching Assistant

- Developed computational frameworks to automate the construction of physics-based models that predict experimental observables. Combined reinforcement learning (PPO) with high-performance computing tools (Python, Mathematica, Slurm) to efficiently explore large model spaces, optimize parameter sets, and achieve higher predictive accuracy with minimal input complexity.
- Authored multiple peer-reviewed publications in theoretical and computational physics, including one integrating AI-driven methods for automated model discovery.
- Delivered instruction and mentorship across 10+ courses from undergraduate to PhD level, developing strong communication and leadership skills while simplifying complex topics for diverse audiences.

National Autonomous University of Mexico: Mexico City, Mexico **2020-2021**

Research Assistant & Teaching Assistant

- Applied numerical and symbolic modeling (Python, Mathematica) to simulate and analyze physical systems. Taught and graded advanced undergraduate courses.

Selected projects

AI-assisted Neutrino Flavor Theory Design **-** University of California: Irvine **2025**

- Developed the "Autonomous Model Builder (AMBer)" framework, a reinforcement-learning system built with Stable Baselines and integrated into a custom end-to-end software pipeline to automate physics-model generation, parameter optimization, and validation against experimental data. Enabled large-scale exploration of theoretical spaces and reduced manual analysis time.
- Created and maintained FlavorBuilder, a Python package supporting model construction, parameter optimization, and data-driven validation.
- Tech stack: Python, PyTorch, Stable Baselines, OpenAI Gym, reinforcement learning (PPO), data analysis, high-performance computing (Slurm).

Text Reliability Classification with Fine-Tuned LLMs - Erdos Institute ML Project **2025**

- Fine-tuned a BERT model for multi-class truthfulness prediction on a Kaggle dataset, benchmarking against a TF-IDF + logistic regression baseline to quantify transformer advantages.
- Tech stack: Python, PyTorch, Hugging Face Transformers, scikit-learn, Pandas, NumPy.

Selected publications and awards

- J. B. Baretz, M. Fieg, V. Ganesh, A. Ghosh, V. Knapp-Perez, J. Rudolph and D. Whiteson, "Towards AI-assisted Neutrino Flavor Theory Design", [arXiv:2506.08080 [hep-ph]]. **2025**
- Latino Excellence and Achievement Award (LEAD), honored for research excellence and leadership of Hispanic/Latinx students at UCI. **2024**
- UC President's Lindau Nobel Laureate Meetings Fellowship, selected to engage with Nobel Laureates and international scientists. **2024**
- UCI Department of Physics and Astronomy Outstanding Graduate Student Award, recognizing top first-year graduate student. **2022**